

Chapter 6

2. Explain the difference between method declaration and method body.

A method declaration defines the first line of a method whereas a method body is the code that runs inside of the curly bracket. The method declaration has the method name, return type and parameters, while the method body contains everything inside the curly brackets.

3. What type of keyword is used to change the access level of a method?

The type of keyword that is used to change access level is an access modifier.

4. What is another word for describing the access level of a method?

Another word for describing the access level of a method is visibility.

5. Explain the scope of each of the variables in the code below:

```
Public class ScopeExample {  
    Public static void method() {  
        int var3;  
        for (int var4 = 0; var4 < 2; var4++) {  
            var3 += 1;  
        }  
    }  
}
```

var4 is only inside the for loop. This means that var4 cannot be used anywhere outside of it since it only exists in the for loop. However with var3, it is declared in the entire method. This means that var3 can be used anywhere inside the method.

6. Write a method declaration for each of the following descriptions:

- a) A class method named `getVowels` that can be called by any other method, requires a **String** parameter, and returns an integer value.

`public static int getVowels (String str)`

- b) A class method named `extractDigit` that can be called by any other method, requires an integer parameter, and returns an integer value

`public static int extractDigit (int num)`

- c) A class method named `insertString` that can be called by any other method, requires an **String** parameter and an integer parameter, and returns a string parameter.

`public static String insertString (String str, int num)`

7.

a) How does the compiler distinguish one method from another?

The compiler distinguishes methods using method signature. This includes the method name and parameter list.

b) Can two methods in the same class have the same name? Explain.

Two methods in the same class can have the same name. This is called method overloading. It works if the parameter list is different.

8.

a) What is the return statement for?

The return statement is used to send a value back to the calling method.

b) How many values can a return statement send back to the calling statement?

A return statement can only return one value.

c) How is the declaration of a method returning a value different from the declaration of a method that does not return a value?

A method that uses a returning value needs to have a return type such as int or string and needs to have a return statement. A method that does not return a value has void and doesn't return anything.

9. Find and explain the error in the code below:

```
public class MethodCallExample {  
  
    public static int doSomething() {  
        return(5);  
    }  
}  
  
public static void main(String[] args) {  
    int num;  
    doSomething();  
    num =doSomething();  
}
```

The main method is outside of the class. This is wrong because you cannot use main after it is closed with the curly bracket of the class. Another error in this code is that syntax for (string[]args) is wrong. You cannot have curly brackets inside. Instead you need square brackets [].

11. Determine if each of the following are true or false. If false, explain why.

a) Breaking a task down into methods is called procedural abstraction.

True

b) A method call consists of the method declaration in an assignment statement.

This is false because the method call uses the method name, not the declaration.

c) A void method must return a value.

This is false because the void method returns nothing

d) An access modifier declares the return type of a method.

This is false because it controls the visibility.

e) The keyword static declares a method is a class method.

True

f) Method parameters are enclosed by braces ({}).

This is false. It uses parentheses ().

g) Local variables can be used by any method in a class

This is false because it is only within their method.

h) The value of an argument passed to a method can be changed in an assignment statement in the method.

This is false because Java uses pass-by-value.

i) Method overloading means that an application contains more than 10 methods.

This is false because it needs to use different parameters in order to work. Otherwise, it would work.

j) The return statement is used to send a value back to the calling statement

True

k) The precondition of a method states the data types of the method's parameters.

This is false because the precondition describes what is true before the method runs.

l) The postcondition of a method describes the way the method accomplishes its task.

This is false because it declares the result, but not the process.